DATA ENGINEERING PROJECT

POWER BI

ANALYTICS AND REPORTING

Enterprise Data Ingestion, Data Modeling, DAX Analytics and Interactive Reporting

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Project Implementation Documentation

Abstract

This project focuses on developing a Power BI analytics workflow for sales and customer reporting. The implementation covers data ingestion from CSV files, Excel workbooks, and SQL Server, followed by data preparation, relational modeling, analytical measure development, and interactive visualization.

Microsoft SQL Server was used for the local database `PowerBI_Engineering`. Power Query was used to prepare the source data, and Power BI Desktop was used to create the analytical model and DAX calculations.

The model contains the sales fact table `FactInternetSales` and the dimensions `DimDate`, `DimCustomer`, `DimProduct`, and `DimTerritory`. A centralized `_Measures` table was created to organize measures for revenue, cost, gross margin, margin percentage, ship-date sales, year-to-date revenue, prior-year sales, year-over-year growth, rolling twelve-month sales, and customer ranking.

The reporting work includes customer-level analysis, revenue analysis, a decomposition tree, and time-intelligence visuals. The final capstone design also covers machine-learning output integration, Power BI Service publishing, scheduled refresh, gateway configuration, and row-level security.

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1 Project Overview

1.1 Introduction

Business intelligence enables organizations to transform operational data into meaningful information for decision-making. Reliable reporting requires data to be extracted, cleaned, validated, structured, and presented in an understandable format.

This project applies Microsoft Power BI to a retail sales analytics scenario. The workflow includes preparing data from multiple sources, creating relationships between fact and dimension tables, developing reusable DAX measures, and constructing interactive report visuals.

SQL Server provides the relational database environment used during the practical implementation. Power Query supports data preparation, while Power BI Desktop supports data modeling, calculations, and visualization.

1.2 Objectives

The major objectives of the project are:

- Ingest data from CSV files, Excel workbooks, and SQL Server.
- Clean and transform data using Power Query.
- Apply appropriate data types to date, key, text, and numeric columns.
- Prepare fact and dimension tables for analytical reporting.
- Establish one-to-many relationships with single-direction filtering.
- Configure active and inactive date relationships.
- Create reusable DAX measures in a centralized table.
- Implement profitability, ranking, and time-intelligence calculations.
- Develop interactive visuals for sales and customer analysis.

- Document the requirements for deployment and data governance.

1.3 Technology Stack

Technology	Purpose
Microsoft Power BI Desktop	Data modeling, DAX calculations, and report visualization.
Power Query	Data ingestion, transformation, cleaning, and type conversion.
Microsoft SQL Server	Relational database and storage for project tables.
SQL Server Management Studio	SQL execution, table inspection, and data validation.
DAX	Business calculations and time-intelligence measures.
Power BI Service	Target environment for publishing, refresh, and report sharing.

1.4 Database Environment

The local SQL Server database used during the practical work was named `PowerBI_Engineering`.

The principal analytical tables are:

- `FactInternetSales`
- `DimDate`
- `DimCustomer`
- `DimProduct`
- `DimTerritory`

Additional source queries include `Sales_Orders` and `MonthlyForecast`.

The database was modified and populated during the practical sessions. Locally generated test records should be distinguished from original Adventure Works records when documenting data provenance.

2 Exercise 1: Data Ingestion and Power Query ETL

2.1 Objective

The objective was to prepare a data ingestion workflow using different source systems. The exercise covered folder-based CSV ingestion, Excel customer data, SQL Server product data, source parameters, schema enforcement, and forecast transformation.

Power Query was used to inspect the source structures and apply transformations before loading prepared data into the Power BI model.

2.2 Source Ingestion

Entity	Source	Purpose
Sales_Orders	CSV folder	Combine files, promote headers, parse dates, and clean invalid rows.
DimCustomer	Excel workbook	Prepare customer names and clean customer text fields.
DimProduct	SQL Server	Retrieve product attributes and prepare active-product data.
MonthlyForecast	CSV file	Transform monthly forecast columns into a normalized structure.

2.3 Sales Orders Transformation

The `Sales_Orders` query was created using CSV folder ingestion. The transformation process included combining files and promoting the first row to become the column headers.

The preparation included:

- Combining source CSV files.
- Promoting headers to column names.
- Converting `OrderDate` to a date type.
- Removing corrupt or invalid rows encountered during cleaning.
- Removing the unnecessary `Source.Name` column where it was not required.
- Applying suitable data types to the resulting columns.

During the practical session, the Power Query preview showed 121 rows and zero errors after the applied transformations. This describes the observed preview at that time and is not a guarantee of the count for future source files.

2.4 Customer Data Preparation

The `DimCustomer` query was used to prepare customer attributes originating from an Excel workbook.

The first-name and last-name fields were combined into a full-name field. The intended cleaning operations include:

- Combining `FirstName` and `LastName`.
- Renaming the resulting field to `CustomerName`.
- Trimming leading and trailing spaces.
- Cleaning email strings.
- Checking column names and data types.

The name-combination step was performed during the practical work. The final rename and email-cleaning steps should be checked in the final query before being marked as verified.

2.5 SQL Server Product Query

The `DimProduct` query was connected to SQL Server. A native SQL query was inspected, using product attributes such as product key, product name, category, standard cost, and active status.

Listing 2.1: Product query structure

```sql
SELECT
    [ProductKey],
    [ProductName],
    [CategoryName],
    [StandardCost],
    [IsActive]
FROM dbo.DimProduct;
```

The local product table contained ten records during the practical session. A filtered-rows step and a changed-type step were also present in Power Query.

The intended final preparation includes filtering active products and assigning an appropriate numeric or currency type to `StandardCost`.

Query folding must be verified separately through the *View Native Query* option for the applicable transformation steps. Inspecting a native SQL query alone does not establish that every later transformation folds.

2.6 Monthly Forecast Transformation

The `MonthlyForecast` query was included as the forecast source. The intended transformation is to retain identifier columns and unpivot the month columns.

The resulting normalized structure is:

Column	Purpose
Date	Forecast date or month
ForecastAmount	Forecast value for the period

This structure makes forecast data easier to analyze in time-based visuals. The final unpivot output and column types should be checked in Power Query before the transformation is considered complete.

2.7 Parameters and Helper Queries

The Power Query environment included the parameters `FilePath` and `ServerName`. These parameters are intended to simplify source management when moving between environments.

Helper queries included:

- `Sample File`
- `Parameter1 (Sample File)`
- `Transform File`
- `Transform Sample File`

These queries support the folder-combination workflow. The `FilePath` and `ServerName` parameter queries were excluded from model loading, while helper queries were retained as needed by the ingestion process.

2.8 Exercise 1 Outcome

The folder-based sales ingestion, basic sales transformations, customer name combination, SQL Server product connection, and parameter configuration were worked on during the practical implementation.

Final verification is still needed for the customer cleaning steps, forecast unpivot output, and query-folding status across the final SQL transformation chain.

3 Exercise 2: Dimensional Modeling and Star Schema

3.1 Objective

This exercise focused on separating transaction-level metrics from descriptive attributes and organizing the model around a sales fact table and supporting dimensions.

A star schema simplifies filtering and makes analytical calculations easier to manage. The fact table contains sales records, while dimension tables provide descriptive attributes.

3.2 Fact Table

The `FactInternetSales` table is the central sales fact table. The fields used during modeling include:

- `SalesAmount`
- `TotalProductCost`
- `OrderDateKey`
- `ShipDateKey`
- `CustomerKey`
- `ProductKey`
- `TerritoryKey`

These fields support sales aggregation and relationships with the associated dimensions.

3.3 Dimension Tables

DimCustomer: Contains customer identifiers, names, and descriptive customer attributes.

DimProduct: Contains product identifiers, names, categories, standard costs, and active status.

DimTerritory: Contains geographic attributes such as region, country, and group.

DimDate: Contains date keys and calendar attributes used for time-based analysis.

3.4 Date Dimension Creation and Validation

The local date dimension was prepared in SQL Server for 1 January 2023 through 31 December 2026.

The validation query was:

Listing 3.1: Validate date dimension

```
USE PowerBI_Engineering;

SELECT
    COUNT(*) AS TotalRows,
    MIN([Date]) AS StartDate,
    MAX([Date]) AS EndDate
FROM dbo.DimDate;
```

The observed validation result was:

Field	Result
TotalRows	1,461
StartDate	2023-01-01
EndDate	2026-12-31

The date dimension includes `DateKey`, `Date`, `FiscalYear`, `Quarter`, and `Month`. The table was also marked as the official date table in Power BI using its date column.

The date range describes the available calendar dimension. It does not prove that sales transactions exist for every date in the range.

3.5 Relationship Configuration

The following principal relationships were established in the model:

Dimension	Fact Column	Status
DimProduct[ProductKey]	FactInternetSales[ProductKey]	Active
DimCustomer[CustomerKey]	FactInternetSales[CustomerKey]	Active
DimTerritory[TerritoryKey]	FactInternetSales[TerritoryKey]	Active
DimDate[DateKey]	FactInternetSales[OrderDateKey]	Active
DimDate[DateKey]	FactInternetSales[ShipDateKey]	Inactive

The intended cardinality is one-to-many $(1 : *)$, with single-direction filtering from the dimensions to the fact table.

The order-date relationship is active. The ship-date relationship is inactive so that a measure can selectively activate it using `USERELATIONSHIP`.

3.6 Modeling Principles

- Use dimension tables to filter and describe sales records.
- Keep relationships at the correct cardinality.
- Prefer single-direction filtering for the star schema.
- Use the order-date relationship as the default date path.
- Activate the ship-date relationship only for the calculations that require it.
- Avoid unnecessary calculated columns in the fact table.

3.7 Exercise 2 Outcome

The core relationships between the fact table and customer, product, territory, and date dimensions were configured. The date table was validated in SQL Server and marked as the date table in Power BI.

The final model should be checked for unintended relationships, key mismatches, and ambiguous filtering paths before submission.

4 Exercise 3: DAX Measures and Time Intelligence

4.1 Objective

This exercise focused on reusable DAX measures for sales, costs, profitability, time-based analysis, and customer ranking. The measures were organized in a separate table named `_Measures` instead of being placed directly inside the fact table.

4.2 Centralized Measures Table

The following expression was used to create the measures table:

Listing 4.1: Create measures table

```
_Measures = DATATABLE("Dummy", INTEGER, {{1}})
```

This creates a small table that acts as a dedicated location for measures. It is disconnected from the business model. The `Dummy` column is a technical placeholder.

4.3 Total Revenue

Listing 4.2: Total Revenue

```
Total Revenue =
SUM(FactInternetSales[SalesAmount])
```

This measure calculates total sales revenue for the records included in the current filter context.

4.4 Total Cost

Listing 4.3: Total Cost

```
Total Cost =
SUM(FactInternetSales[TotalProductCost])
```

This measure calculates the total product cost associated with the filtered sales records.

4.5 Gross Margin

Listing 4.4: Gross Margin

```
Gross Margin =
[Total Revenue] - [Total Cost]
```

Gross margin is calculated by subtracting total product cost from total revenue.

4.6 Margin Percentage

Listing 4.5: Margin Percentage

```
Margin % =
DIVIDE(
    [Gross Margin],
    [Total Revenue],
    0
)
```

The `DIVIDE` function handles a zero denominator by returning the specified alternate result.

4.7 Sales by Ship Date

Listing 4.6: Sales by Ship Date

```
Sales by Ship Date =
CALCULATE(
    [Total Revenue],
    USERELATIONSHIP(
        FactInternetSales[ShipDateKey],
        DimDate[DateKey]
    )
)
```

This measure activates the inactive ship-date relationship for the calculation. It allows revenue to be analyzed using shipment dates rather than order dates.

4.8 Revenue Year to Date

Listing 4.7: Revenue YTD

```
Revenue YTD =
TOTALYTD(
    [Total Revenue],
    DimDate[Date]
)
```

This measure calculates cumulative revenue from the beginning of the applicable year through the date in the current context.

4.9 Prior Year Sales

Listing 4.8: Prior Year Sales

```
Prior Year Sales =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(DimDate[Date])
)
```

This measure evaluates revenue for the corresponding period in the previous year.

4.10 Year-over-Year Growth

Listing 4.9: YoY Growth Percentage

```
YoY Growth % =
DIVIDE(
    [Total Revenue] - [Prior Year Sales],
    [Prior Year Sales],
    0
)
```

The measure calculates the percentage change between current revenue and prior-year revenue.

4.11 Rolling Twelve-Month Sales

Listing 4.10: Rolling 12M Sales

```
Rolling 12M Sales =
CALCULATE(
    [Total Revenue],
    DATESINPERIOD(
        DimDate[Date],
        MAX(DimDate[Date]),
        -12,
        MONTH
    )
)
```

This measure calculates revenue over a rolling twelve-month period ending at the maximum date in the current context.

4.12 Customer Rank

Listing 4.11: Customer Rank

```
Customer Rank =
RANKX(
    ALL(DimCustomer),
    [Total Revenue],
    ,
    DESC,
    DENSE
)
```

This measure ranks customers by revenue in descending order. Dense ranking does not skip rank numbers after ties.

4.13 Measure Summary

Measure	Purpose
Total Revenue	Calculate total sales value.
Total Cost	Calculate total product cost.

Gross Margin	Calculate revenue less cost.
Margin %	Calculate gross margin as a percentage of revenue.
Sales by Ship Date	Analyze revenue using shipment dates.
Revenue YTD	Calculate cumulative revenue within the year.
Prior Year Sales	Calculate revenue for the prior-year period.
YoY Growth %	Calculate percentage change from prior-year sales.
Rolling 12M Sales	Calculate revenue for a rolling twelve-month window.
Customer Rank	Rank customers according to revenue.

4.14 Formatting and Validation

The intended number formats are:

- Currency: `$ #,0`
- Percentage: `0.0%`
- Integer: `#,0`

Formatting should be applied through the measure properties in Power BI Desktop. A complete validation should test the measures under different date, customer, product, and territory filters.

The time-intelligence calculations depend on a correctly configured date table and relationships. Their results should be verified against appropriate date ranges and source data.

4.15 Exercise 3 Outcome

Ten principal measures were created in the centralized `_Measures` table. These measures cover base aggregations, profitability, ship-date analysis, time intelligence, and customer ranking.

The measure library was established. Final formatting and context-based validation should be completed before the measure library is considered fully audited.

5 Exercise 4: Interactive Report Development

5.1 Objective

This exercise focuses on presenting the analytical model through readable, interactive report pages. The design requirements include KPI cards, sales analysis, a decomposition tree, customer drill-through, and report-page tooltips.

5.2 Executive Overview

The executive overview is intended to present the main performance indicators:

- Total Revenue
- Gross Margin
- YoY Growth Percentage

Additional visuals can present revenue by category, region, and time period. A consistent layout and appropriate number formats help users interpret the results.

5.3 Decomposition Tree

A decomposition-tree visual was configured with `Total Revenue` as the analysis measure. The explanation fields included `CategoryName` and `Region`, with a regional field used for tooltip information.

The purpose of the decomposition tree is to explore how revenue varies across product categories and geographic regions. The final interaction should be checked to ensure that users can expand the appropriate branches and interpret the results correctly.

5.4 Customer 360 Page

A report page named `Customer 360` was created for customer-level analysis.

The customer table contained:

- CustomerKey
- CustomerName
- Customer Rank
- Total Revenue

The table showed customer-level revenue and ranking. The displayed total revenue during the local practical session was 19,180.00.

The drill-through configuration used `CustomerKey` as the customer identifier and retained the *Keep all filters* option.

The intended interaction is to select a customer from a source visual and navigate to the Customer 360 page with the selected customer's filter context. The final drill-through behavior should be tested from the source visual.

5.5 Time-Intelligence Visuals

A line chart was created to examine the `Rolling 12M Sales` measure over time. The report also included a visual for `Revenue YTD`.

These visuals are intended to help users understand cumulative revenue and longer-term trends. The displayed values should be checked against the date context and underlying sales data.

5.6 Report-Page Tooltip

The project design calls for a compact report-page tooltip that presents trend information when a user hovers over a visual data point.

The work included a rolling twelve-month line chart and a Revenue YTD card. The tooltip page size and its connection to the source visual must be verified to establish that the hover interaction is fully operational.

5.7 Design Principles

The report design requirements include:

- Use a consistent 16:9 canvas for standard pages.
- Place key performance indicators near the top.
- Use readable labels and consistent formatting.
- Avoid unnecessary visual clutter.
- Provide a clear path from summary to detailed analysis.
- Test slicers, cross-filtering, and drill-through behavior.
- Verify the tooltip page configuration.

5.8 Exercise 4 Outcome

The Customer 360 page and customer table were created. The decomposition-tree visual and time-intelligence visuals were also explored.

The final report layout, KPI reference labels, tooltip activation, visual styling, and end-to-end interactions require final verification before the complete report design can be considered finished.

6 Exercise 5: Capstone Deployment and Governance

6.1 Objective

The capstone exercise describes the transition from a local Power BI Desktop model to a governed analytical solution in Power BI Service.

The requirements include machine-learning output integration, report publishing, scheduled refresh, gateway configuration, and row-level security.

These requirements extend beyond local report development and depend on the availability of source files, credentials, permissions, and a working Power BI Service environment.

6.2 Machine-Learning Output Integration

The capstone design calls for integrating customer-level machine-learning predictions. The intended prediction data contains fields such as:

- CustomerKey
- ChurnProbability
- SpendingCluster

The prediction output may be supplied through a CSV file or a supported cloud-storage source. The customer key is intended to associate each prediction with the customer dimension.

The relationship cardinality must be selected according to the actual data. A one-to-one relationship is appropriate only when the key is unique in both tables.

The integration should be validated by checking data types, key coverage, missing customer keys, and duplicate records.

6.3 Churn Risk Categories

The capstone design proposes the following risk categories:

- High risk: probability greater than 0.65.
- Medium risk: probability from 0.35 through 0.65.
- Low risk: probability below 0.35.

These categories can support a risk slicer, customer scatter plot, and high-risk customer analysis. A revenue-at-risk indicator can also be developed after the prediction output has been connected and validated.

6.4 Power BI Service Publishing

The deployment plan includes:

1. Save the completed Power BI Desktop file.
2. Publish the report and semantic model to the designated Power BI Service workspace.
3. Verify that the required data sources and credentials are configured.
4. Test report access and the behavior of published visuals.
5. Package the report for controlled distribution where the workspace and licensing support the required features.

Publishing is considered complete only after the report is available in the intended workspace and its data and visuals have been verified.

6.5 Scheduled Refresh and Gateway

The deployment requirements include configuring scheduled refresh through the appropriate gateway where required by the source environment.

The intended refresh configuration includes a daily refresh at 06:00 UTC. This is a target configuration from the capstone requirements, not confirmation that a scheduled refresh has already been activated.

The final validation should check:

- Gateway installation and availability.
- Mapping of data sources to the gateway.
- Credentials and authentication.
- Successful manual refresh.
- Scheduled refresh configuration.
- Refresh history and error messages.

6.6 Row-Level Security

Row-level security is intended to restrict report data according to the user's permitted region or country.

The capstone design proposes a role named `Regional_Director`. The security logic should associate the signed-in user's email address with the countries that the user is authorized to access.

The following is a conceptual example of the intended lookup logic. The actual column names must match the final model.

Listing 6.1: Illustrative RLS lookup

```
VAR PermittedCountry =
    LOOKUPVALUE(
        UserAccess[PermittedCountry],
        UserAccess[UserEmail],
        USERPRINCIPALNAME()
    )
RETURN
    DimTerritory[Country] = PermittedCountry
```

This example assumes that a single permitted country is returned for each user. If users can access multiple countries, the security expression must be adapted to the access-table design.

The role should be tested with representative user accounts to confirm that each user sees only the permitted data.

6.7 Capstone Outcome

The capstone requirements cover machine-learning data integration, risk-based analytics, publishing, refresh configuration, and row-level security.

These items should be marked complete only after the prediction data has been integrated, the report has been published, scheduled refresh has been tested, and the security role has been validated. Local report development alone does not demonstrate successful service deployment.

7 Validation and Quality Checks

7.1 Data Validation

The quality of the analytical model depends on the quality of the data. The following checks should be included in the final validation process:

- Verify the source and destination row counts.
- Check date, numeric, and key data types.
- Identify missing or invalid keys.
- Check duplicate dimension keys.
- Confirm that fact-table foreign keys match dimensions.
- Inspect transformation errors in Power Query.
- Validate SQL query results independently where possible.

7.2 Model Validation

The model validation should confirm that:

- Dimension keys are unique on the one side of each one-to-many relationship.
- The intended filter direction is single direction.
- The order-date relationship is active.
- The ship-date relationship is inactive.
- The date dimension is marked as the date table.
- Unnecessary relationships do not introduce ambiguous paths.

7.3 DAX Validation

The DAX measures should be evaluated in visuals with appropriate date and dimension fields.

- Compare Total Revenue with the underlying sales values.
- Compare Total Cost with the source cost values.
- Check Gross Margin against Revenue minus Cost.
- Confirm that Margin % behaves correctly when revenue is zero.
- Test Sales by Ship Date using date filters.
- Compare Revenue YTD and Prior Year Sales for suitable periods.
- Check YoY Growth when prior-year revenue is available.
- Validate Rolling 12M Sales over a continuous date range.
- Check customer ranking under different filter contexts.

7.4 Report Validation

The final report should be tested for:

- Correct KPI values and number formats.
- Consistent page layout and readable titles.
- Appropriate interactions between visuals.
- Correct customer drill-through behavior.
- Working report-page tooltips.
- Accurate filtering by date, product, customer, and territory.

7.5 Project Overview

This project involved building a Power BI analytics solution using data ingestion, transformation, dimensional modelling, DAX measures, interactive visualizations, and deployment-related concepts. The implementation used Power BI Desktop, Power Query, SQL Server Management Studio, and a SQL Server database named `PowerBI_Engineering`.

The work focused on preparing data for analysis, organizing fact and dimension tables, creating reusable analytical measures, and presenting business information through report visuals. SQL was used to create and validate database tables, while Power Query was used to inspect source data and manage transformation steps.

The following sections document the five project exercises and include screenshots from the actual implementation.

7.6 Implementation of Exercises 1 to 5

7.6.1 Exercise 1: Data Ingestion and Power Query ETL

Caption: Creating the DimProduct Table

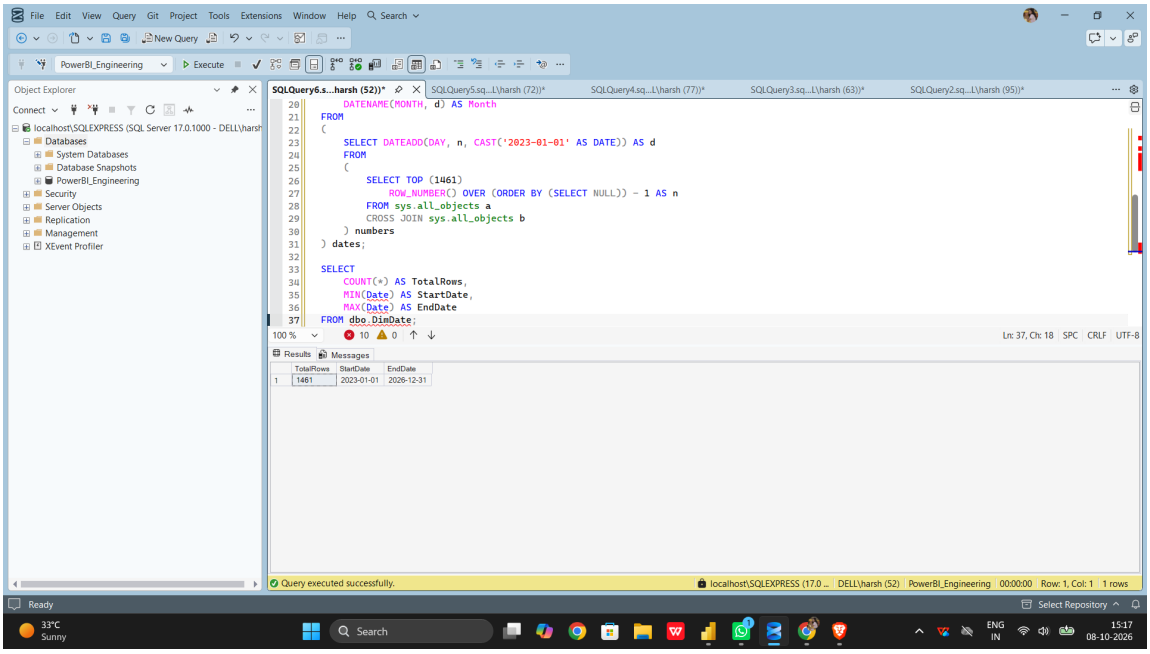


Figure 7.1: Creating the DimProduct Table

This image shows the SQL query used to create the DimProduct table with columns for product key, product name, category, standard cost, and active status. Primary key constraints and appropriate data types are defined to maintain data consistency.

Caption: Power Query Native SQL Query

This image displays the native SQL query used to retrieve product information from the database. The required columns are selected for further transformation and integration into the Power BI data model.

Caption: Date Table Validation

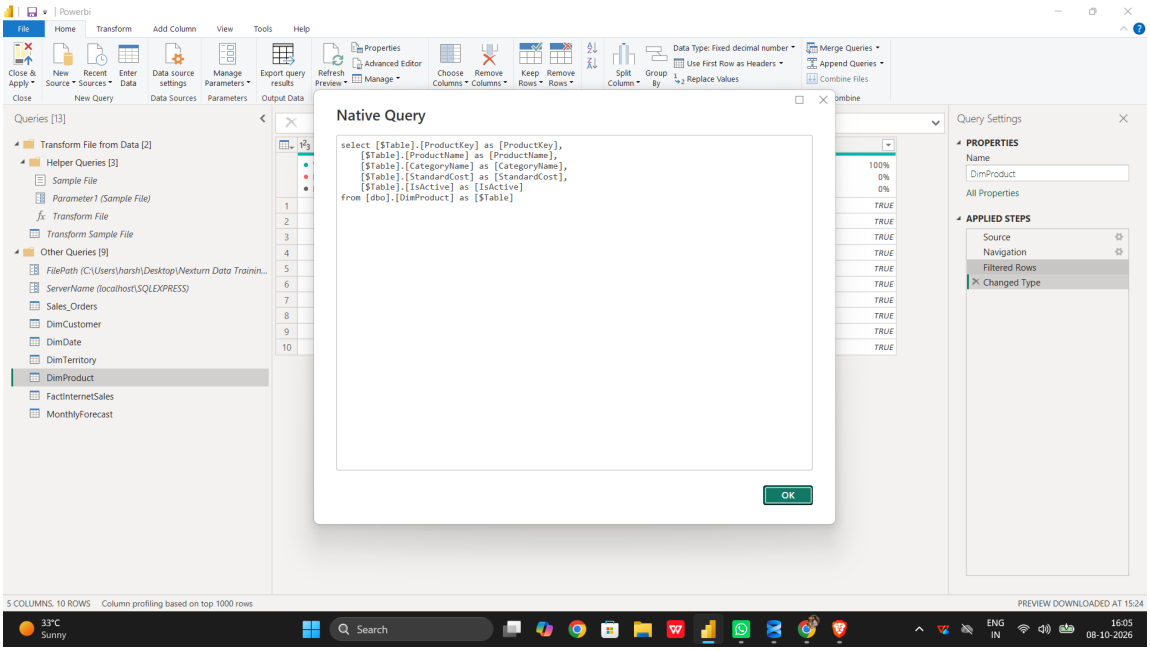


Figure 7.2: Connecting to SQL Server Using Power Query

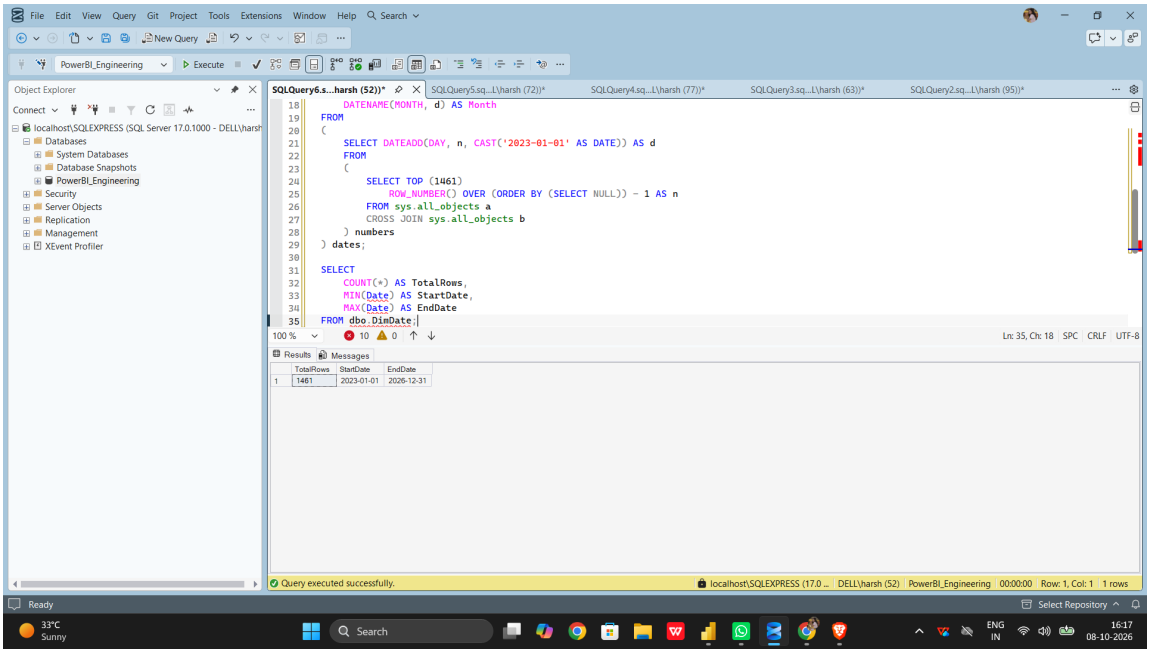


Figure 7.3: Validating the Date Dimension

This image shows the SQL query results used to validate the date dimension. The table contains 1,461 rows, covering the period from 1 January 2023 to 31 December 2026.

7.6.2 Exercise 2: Dimensional Modeling

Caption: Power Query Data Transformation

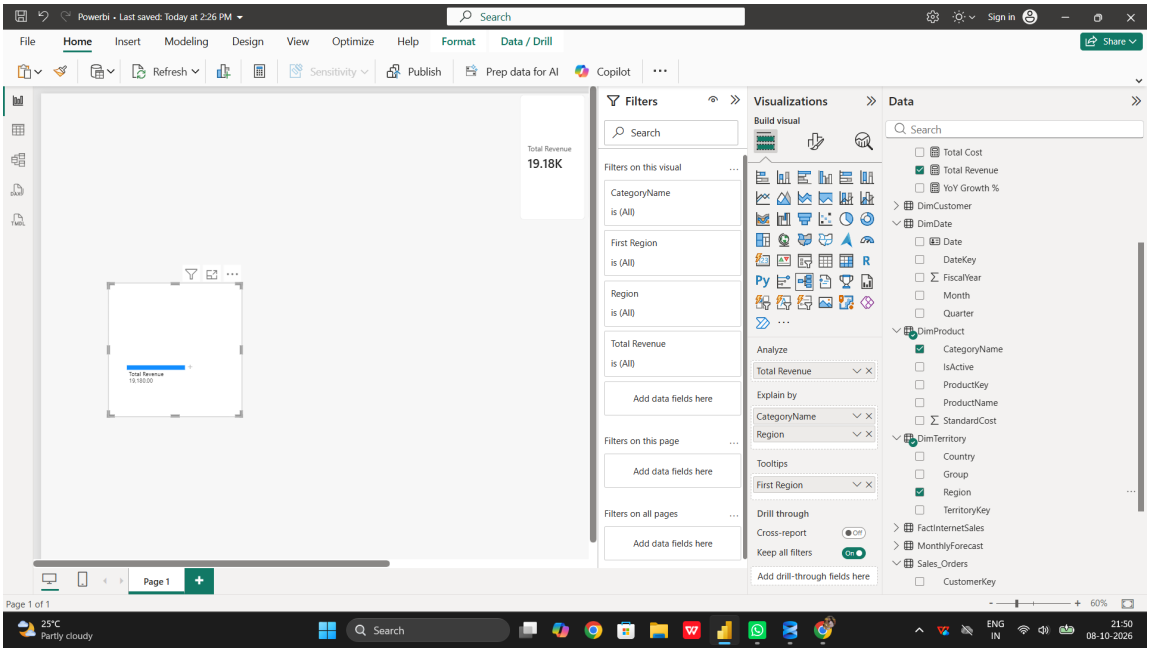


Figure 7.4: Reviewing Data Transformation Steps

This image displays the Power Query Editor with the DimProduct query selected. The applied steps include source selection, navigation, row filtering, and data type conversion to prepare the data for analysis.

Caption: Customer Revenue and Ranking Analysis

This image presents customer identifiers, customer names, customer rankings, and total revenue in a Power BI table. It helps compare customer contributions and identify customers with higher sales values.

7.6.3 Exercise 3: Advanced DAX Measures

Caption: Creating the Total Revenue Measure

This image shows the creation of the Total Revenue measure using the SUM function on the SalesAmount column. The measure calculates total sales and can be reused across different Power BI visuals.

Caption: Rolling Twelve-Month Sales Analysis

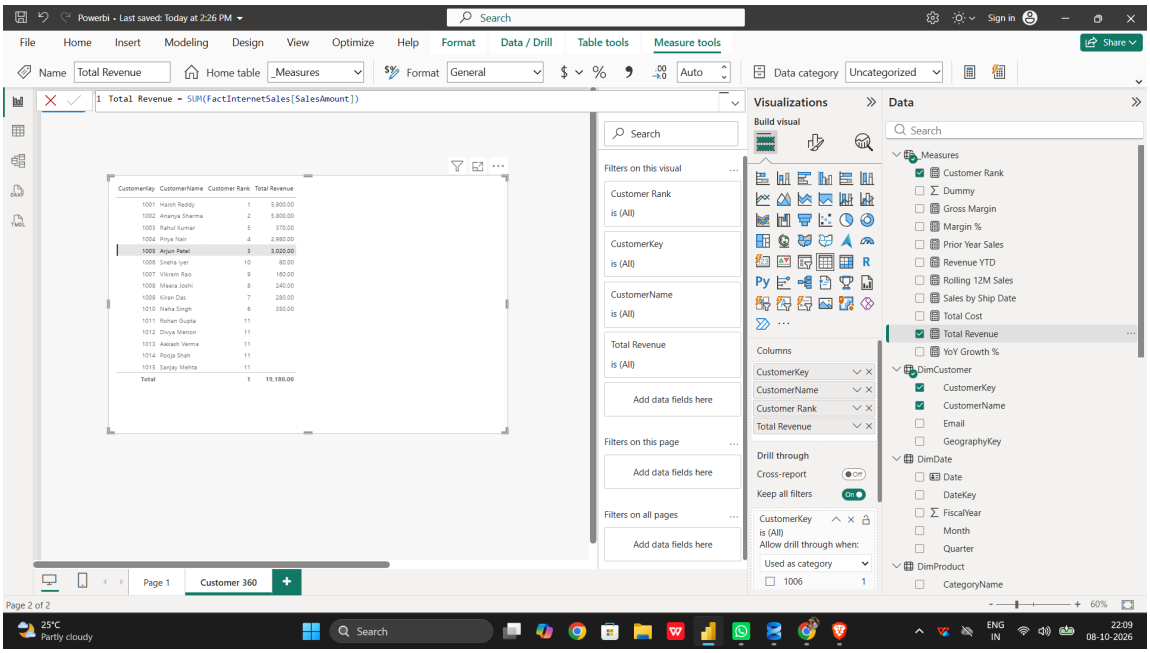


Figure 7.5: Displaying Customer Revenue and Rankings

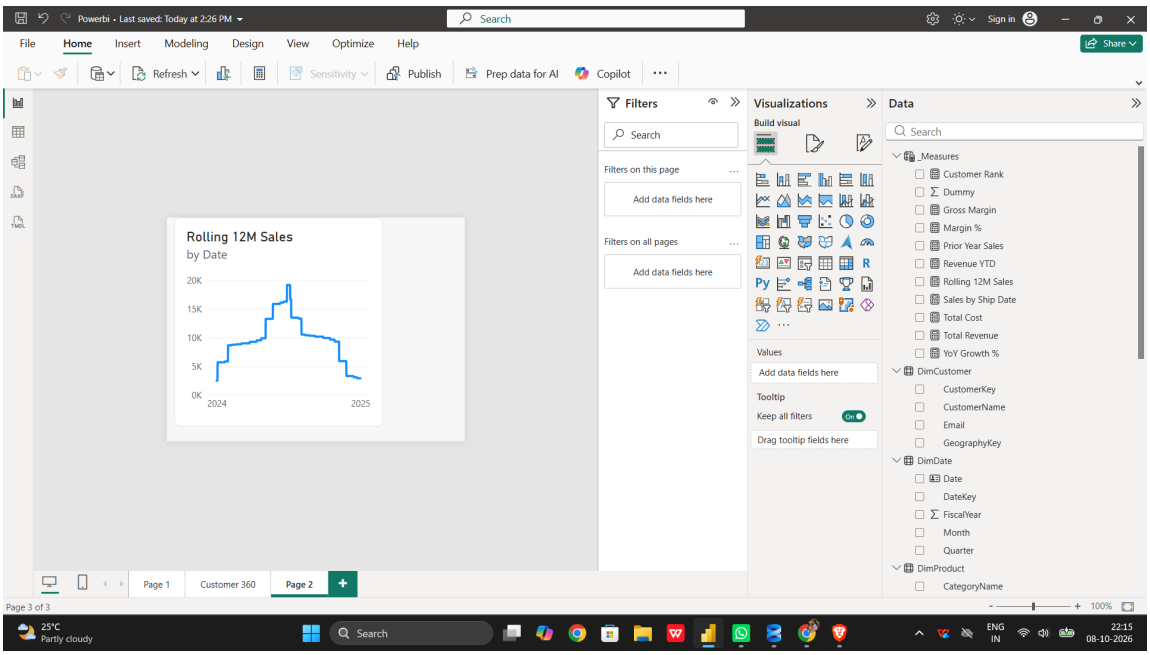


Figure 7.6: Defining the Total Revenue DAX Measure

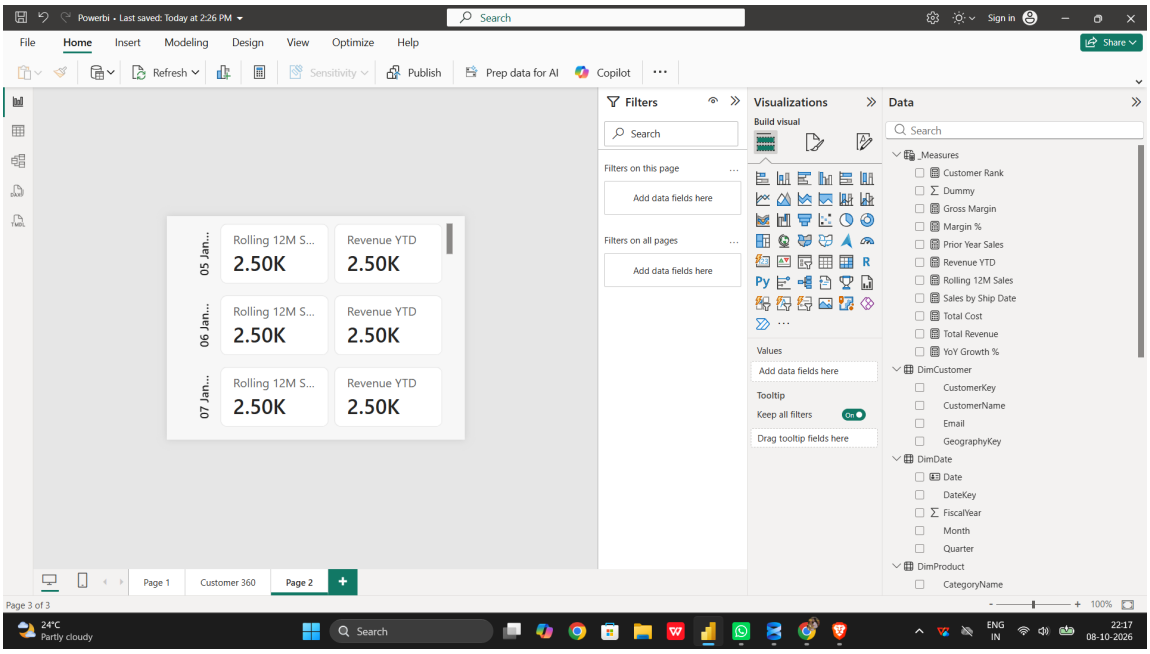


Figure 7.7: Visualizing Rolling Twelve-Month Sales

This image displays a line chart representing rolling twelve-month sales over time. The chart helps identify changes in sales performance and observe revenue trends across the reporting period.

7.6.4 Exercise 4: Report Visualization and Design

Caption: Customer-Level Revenue Report

This image shows a customer-level report containing customer keys, customer names, rankings, and revenue values. The visual supports detailed customer analysis and displays the available drill-through configuration options.

Caption: Revenue Analysis by Category and Region

This image shows the configuration of a revenue analysis visual using Total Revenue as the measure and CategoryName and Region as explanatory fields. These fields support the analysis of revenue contributions across product categories and geographical regions.

7.6.5 Exercise 5: Pipeline Validation

Caption: Validating the Reporting Date Range

This image shows the validation of the date dimension using SQL queries to calculate the total number of records and the minimum and maximum dates. The results confirm the reporting date range and provide a foundation for consistent time-based analysis.

Date	Total Revenue
05 January 2024	2,500.00
10 January 2024	3,200.00
15 February 2024	150.00
02 March 2024	2,800.00
20 March 2024	120.00
12 April 2024	80.00
08 May 2024	160.00
18 June 2024	240.00
22 July 2024	280.00
14 August 2024	350.00
05 September 2024	3,400.00
11 October 2024	2,600.00
Total	**19,180.00**

Figure 7.8: Displaying Customer-Level Revenue Information

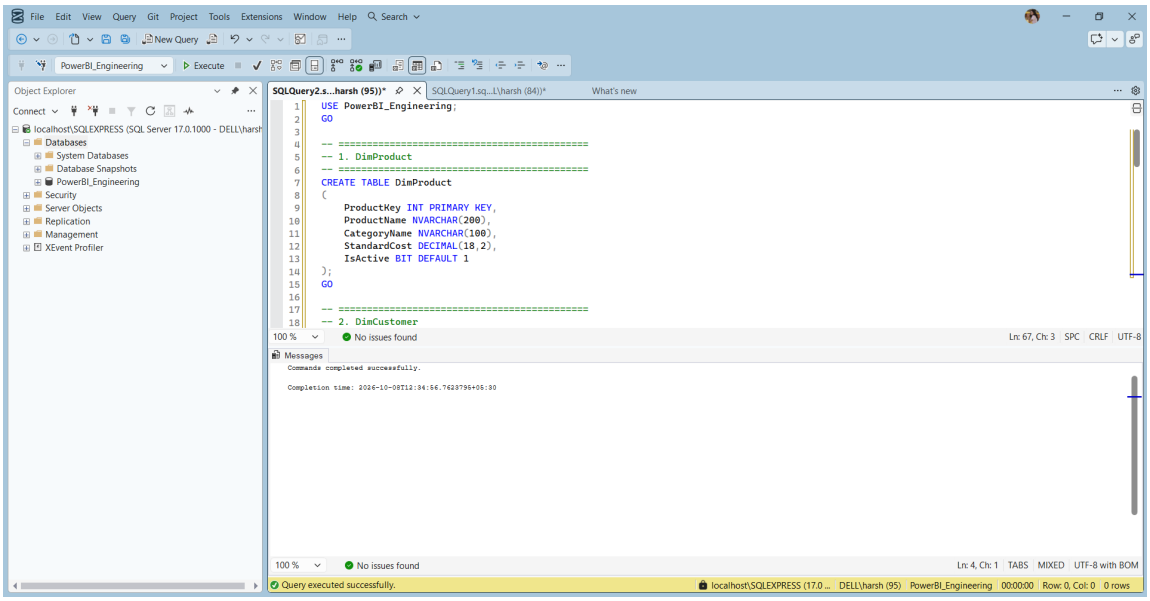


Figure 7.9: Configuring Revenue Analysis

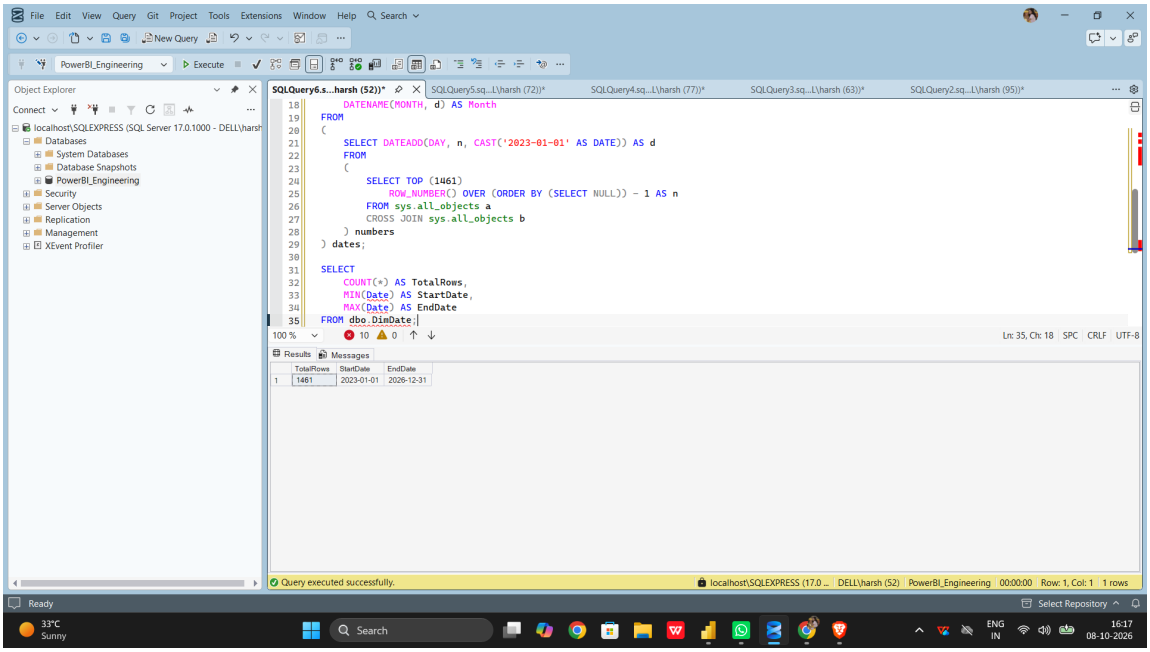


Figure 7.10: Validating Date Coverage for Reporting

7.6.6 Exercise 1: Data Ingestion and Power Query ETL

Caption: Importing Customer Churn Predictions from CSV

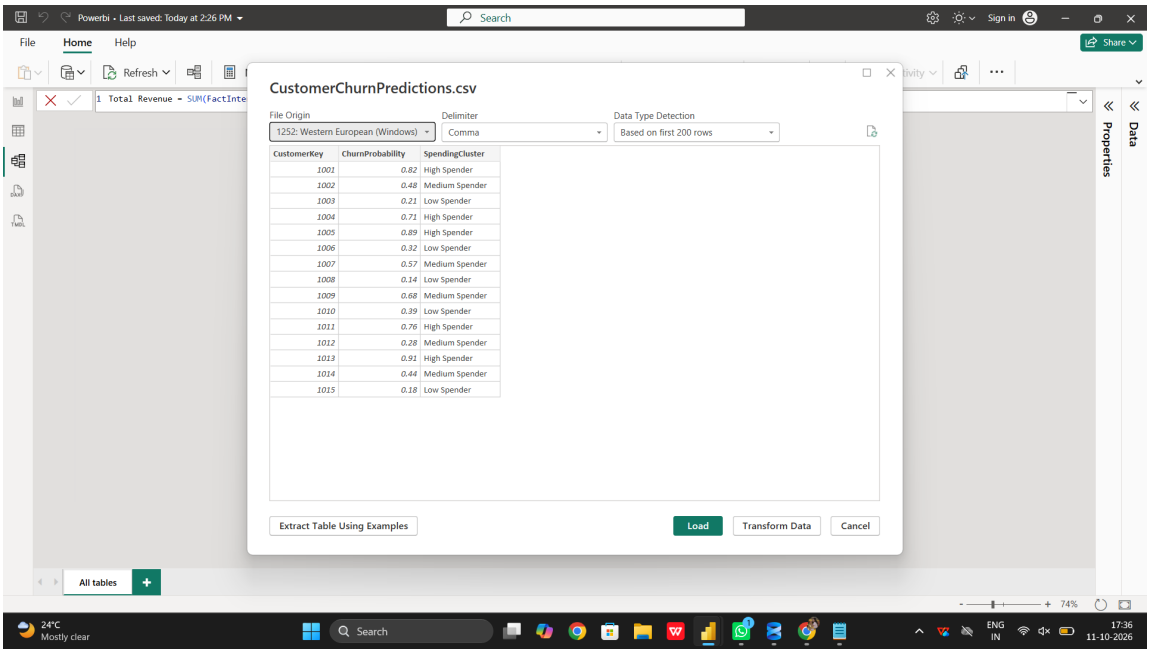


Figure 7.11: Importing Customer Churn Prediction Data

This image shows the import of customer churn prediction data from a CSV file into Power BI. The dataset contains CustomerKey, ChurnProbability, and SpendingCluster columns. The preview displays predicted churn probabilities and customer spending categories, preparing the data for further transformation and analysis.

Caption: Transforming Monthly Forecast Data

This image displays the MonthlyForecast table in Power Query Editor with ProductKey, CategoryName, Date, and ForecastAmount columns. The date column is converted to the date data type, while the forecast amounts are maintained as numeric values. The column profiling indicators show valid values without detected errors or empty entries in the displayed columns.

Caption: Loading Sales Data from SQL Server

This image shows the FactInternetSales table retrieved from SQL Server. It contains sales order numbers, order and shipping date keys, customer keys, product keys, sales amounts, and total product costs. The displayed data confirms that the transactional information is available for integration into the Power BI data model.

Caption: Transforming Product Dimension Data

This image presents the DimProduct table with product identifiers, product names, categories, standard costs, and active status. The StandardCost column is assigned a currency

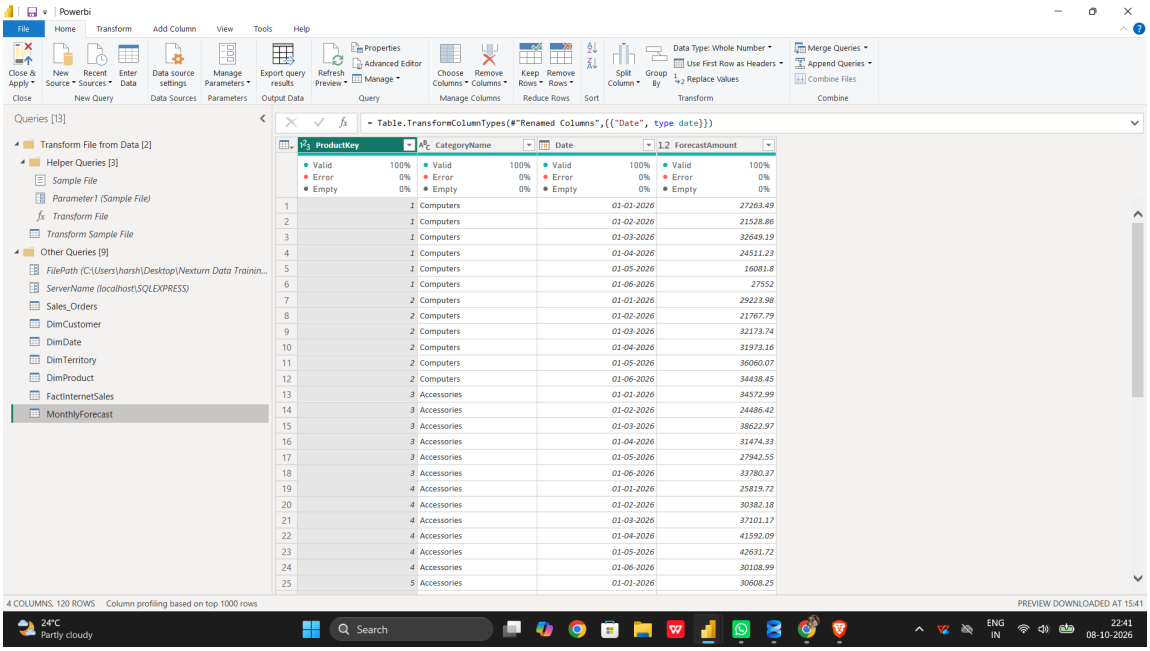


Figure 7.12: Transforming Monthly Forecast Data in Power Query

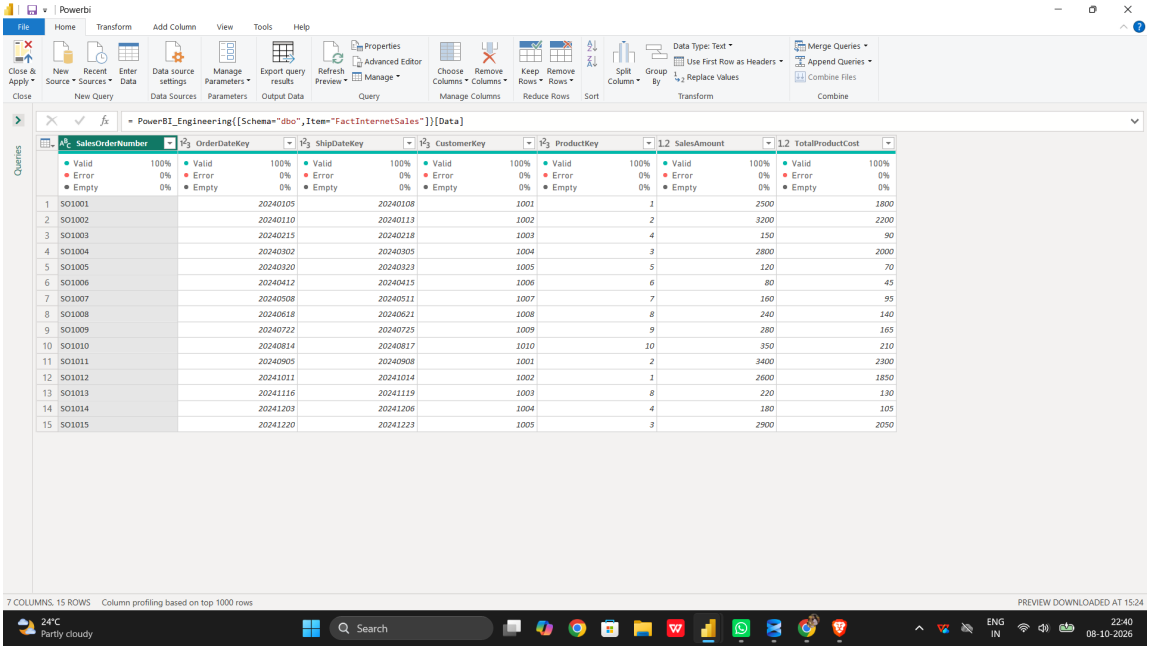


Figure 7.13: Previewing the FactInternetSales Table

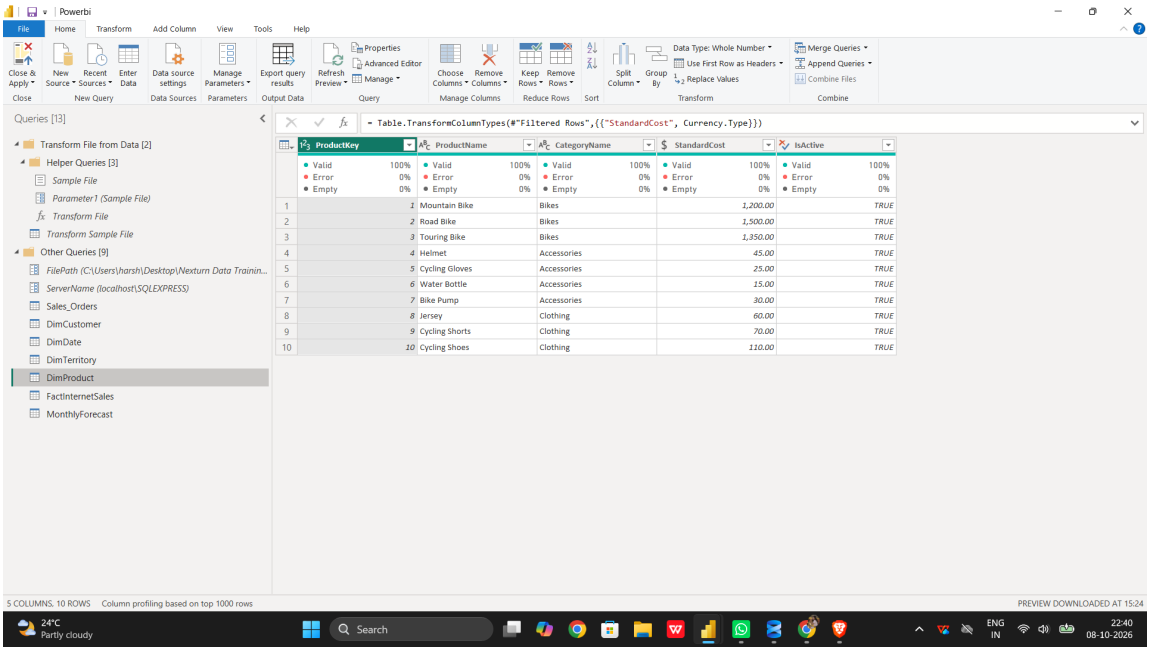


Figure 7.14: Applying Data Types to Product Attributes

data type, while the other columns retain their appropriate types. These transformations help maintain consistency in product information and support accurate cost analysis.

7.6.7 Exercise 2: Dimensional Modeling

Caption: Preparing Territory Dimension Data

This image shows the DimTerritory table containing TerritoryKey, Region, Country, and Group columns. The table organizes geographical information into structured attributes, allowing sales data to be analyzed by region, country, and geographical group. These attributes support geographical filtering and comparison in Power BI reports.

Caption: Creating the Date Dimension

This image displays the DimDate table containing DateKey, Date, FiscalYear, Quarter, and Month columns. The date dimension organizes calendar information into meaningful reporting attributes. These fields support date filtering, quarterly comparisons, and year-based analysis throughout the Power BI report.

Caption: Cleaning Customer Dimension Data

This image presents the DimCustomer table with CustomerKey, CustomerName, Email, and GeographyKey columns. The transformation shown in the formula bar trims unnecessary spaces from email values. This cleaning step improves the consistency of customer contact information and prepares the customer dimension for integration with related sales records.

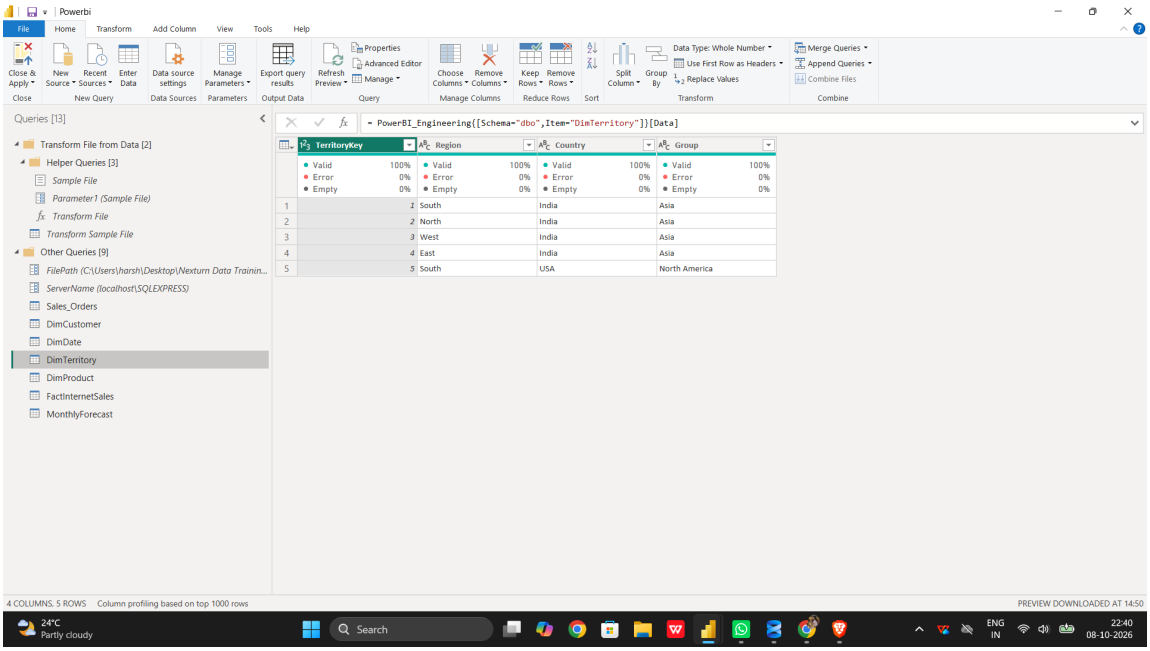


Figure 7.15: Reviewing Regional and Country Information

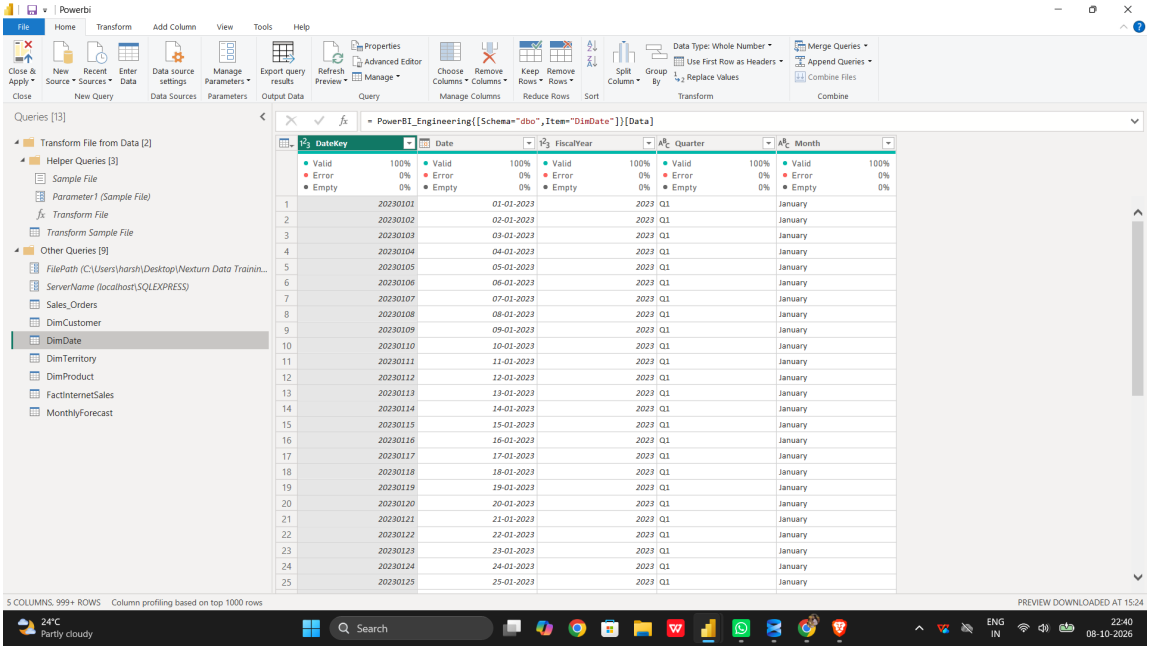


Figure 7.16: Preparing Date Attributes for Time-Based Analysis

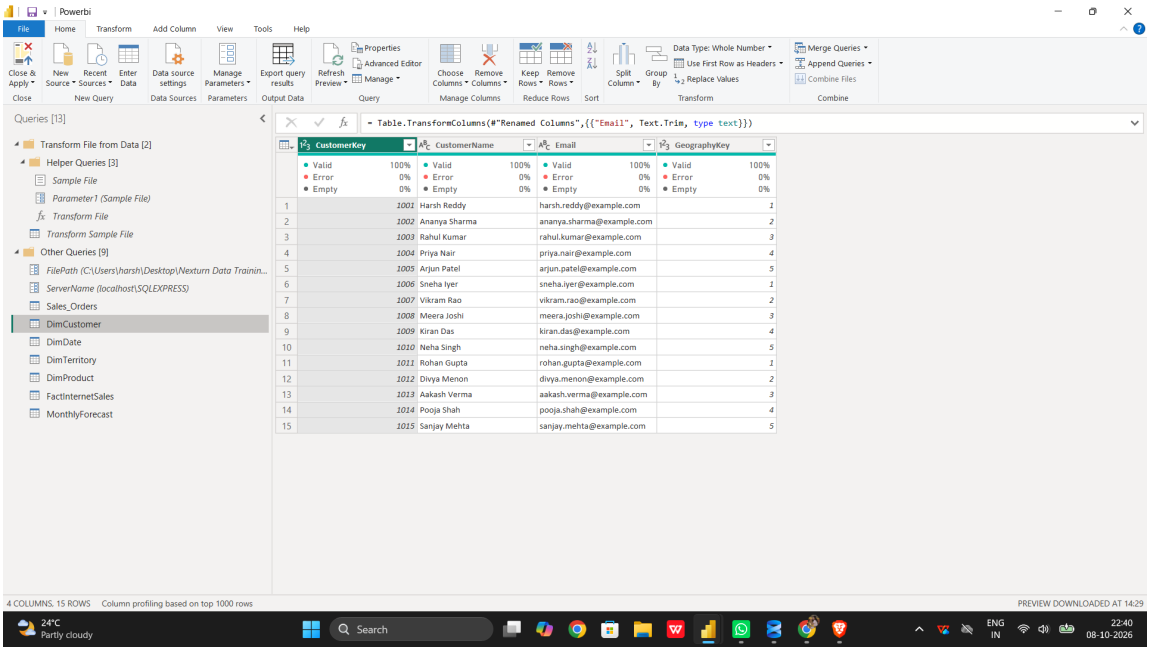


Figure 7.17: Transforming Customer Details in Power Query

7.6.8 Exercise 3: Advanced DAX Measures

Caption: Cleaning and Preparing Sales Order Records

This image shows the Sales_Orders table after a transformation that removes the Source.Name column. The remaining columns include sales order numbers, order dates, shipping dates, customer keys, product keys, territory keys, sales amounts, and total product costs. The cleaned dataset retains the fields needed for sales analysis and reporting.

7.6.9 Exercise 4: Report Visualization and Design

Caption: Establishing Relationships Between Data Tables

This image displays the Power BI model view with FactInternetSales, DimProduct, DimDate, DimCustomer, DimTerritory, MonthlyForecast, Sales_Orders, and the centralized measures table. The relationship lines connect the tables and illustrate how the model is organized for reporting. This structure enables the use of dimension attributes and measures together in analytical visuals.

7.6.10 Exercise 5: Pipeline Validation and Reporting

Caption: Executive Sales Dashboard

This image shows a Power BI report containing KPI cards for Total Revenue, Total Cost, Gross Margin, Margin Percentage, Sales by Ship Date, and Year-over-Year Growth. It

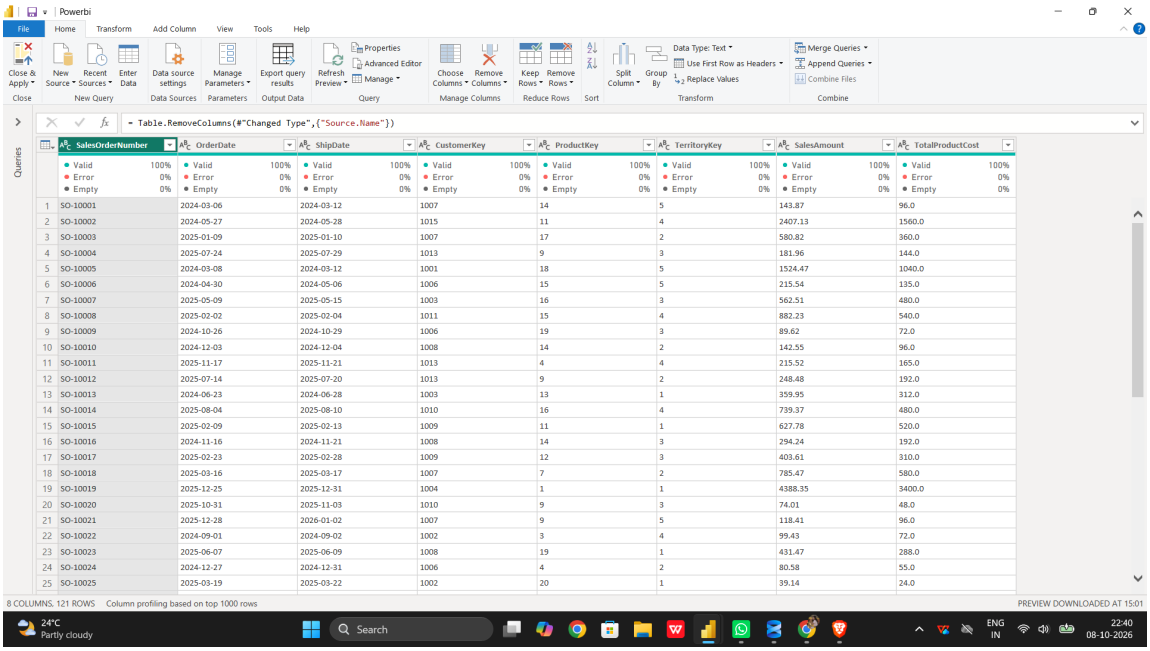


Figure 7.18: Preparing Sales Orders for Analysis

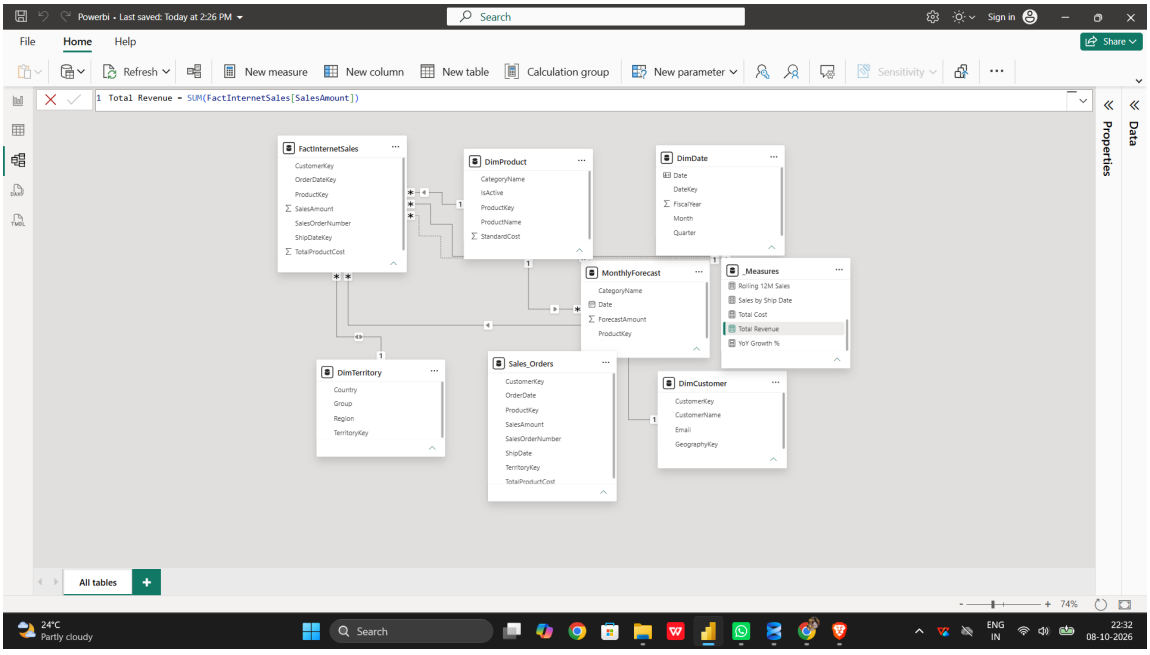


Figure 7.19: Power BI Data Model and Table Relationships

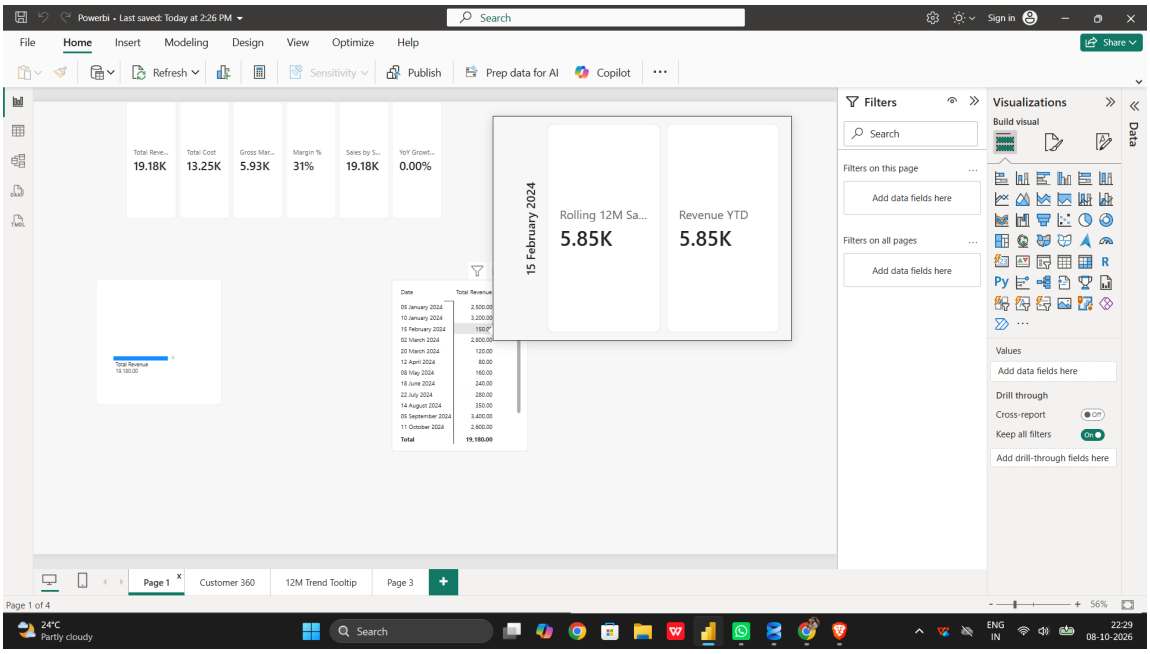


Figure 7.20: Displaying Sales Performance and Time Intelligence Metrics

also includes a revenue table and cards for Rolling 12M Sales and Revenue YTD. These visuals present key sales metrics in a single report layout, helping users review revenue, costs, profitability, and time-based performance.

7.6.11 Exercise 1: Customer Churn Data Preparation

Caption: Customer Churn Risk Analysis Dashboard in Power BI

This image presents the Customer Churn Risk Analysis report created in Power BI. It includes a column chart showing average churn probability by spending cluster, customer count cards, and visual comparisons of customer distribution. The report uses the CustomerChurnPredictions dataset to examine customer churn probabilities and spending behavior. These visualizations help identify customer segments that may require additional attention and support targeted customer retention strategies.

7.6.12 Exercise 2: Data Modeling and Relationship Configuration

Caption: Establishing a Relationship Between Sales and Churn Data

This image shows the relationship configuration between the Sales_Orders and CustomerChurnPredictions tables using the CustomerKey column. The relationship is configured as many-to-one with single-direction cross-filtering and is marked active. This integration allows customer churn information to be analyzed alongside sales transactions, enabling

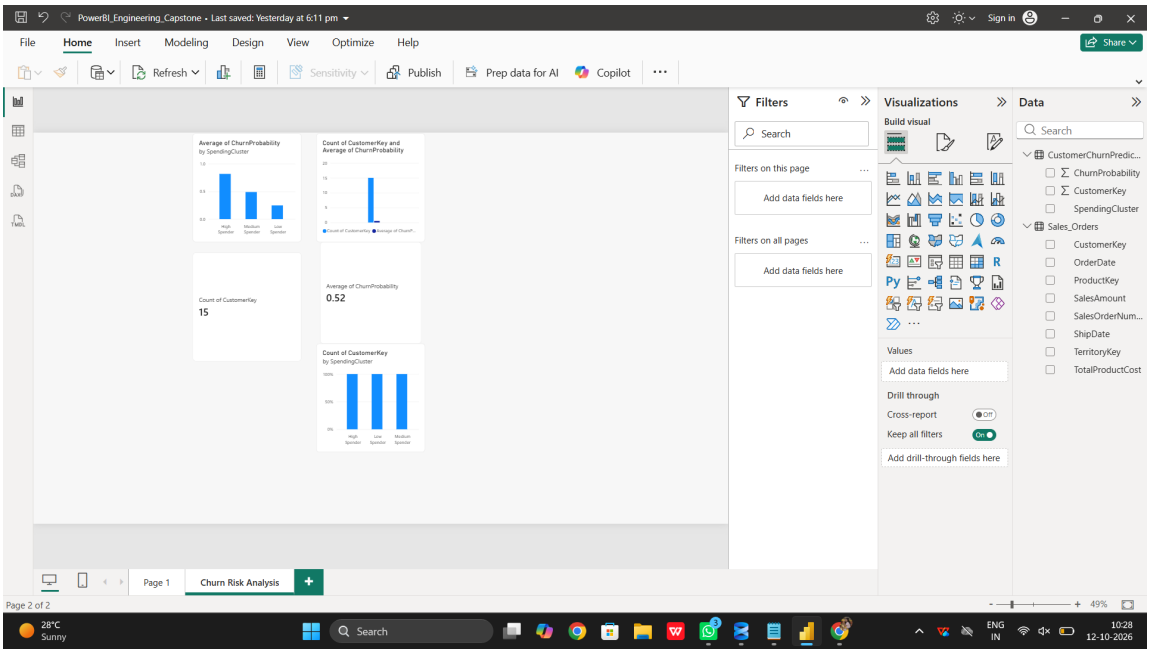


Figure 7.21: Analyzing Customer Churn Probability and Spending Clusters

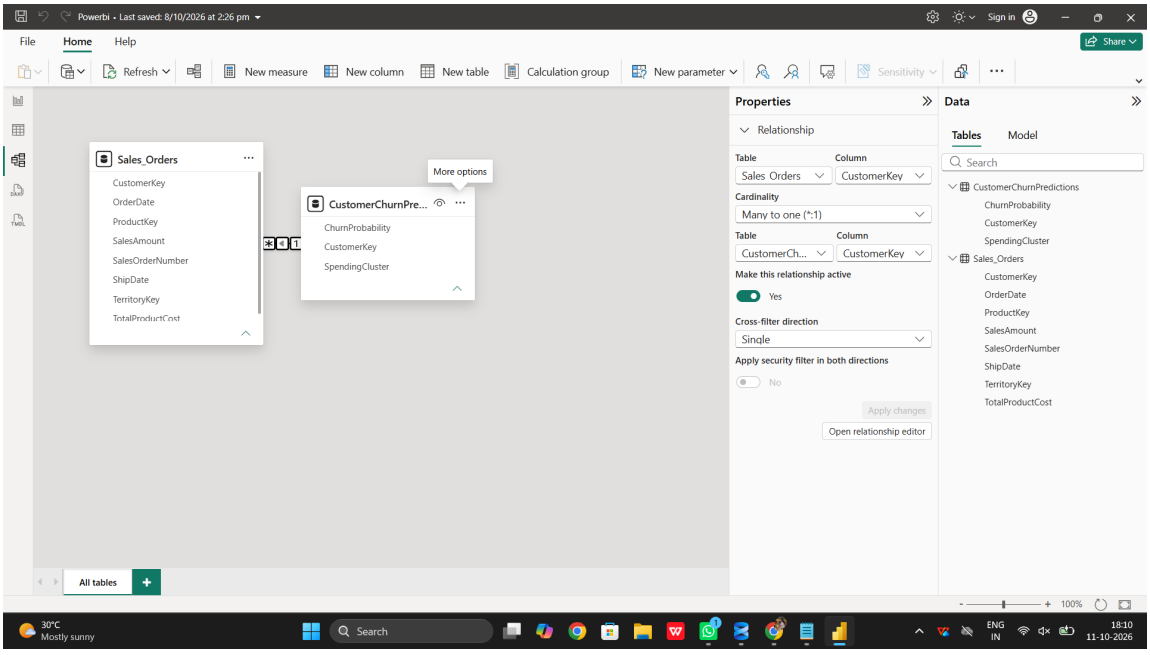


Figure 7.22: Configuring the CustomerKey Relationship

a combined view of customer purchasing activity and churn risk.

7.6.13 Exercise 3: Sales Performance Dashboard

Caption: Creating Sales KPI Cards and Revenue Summary

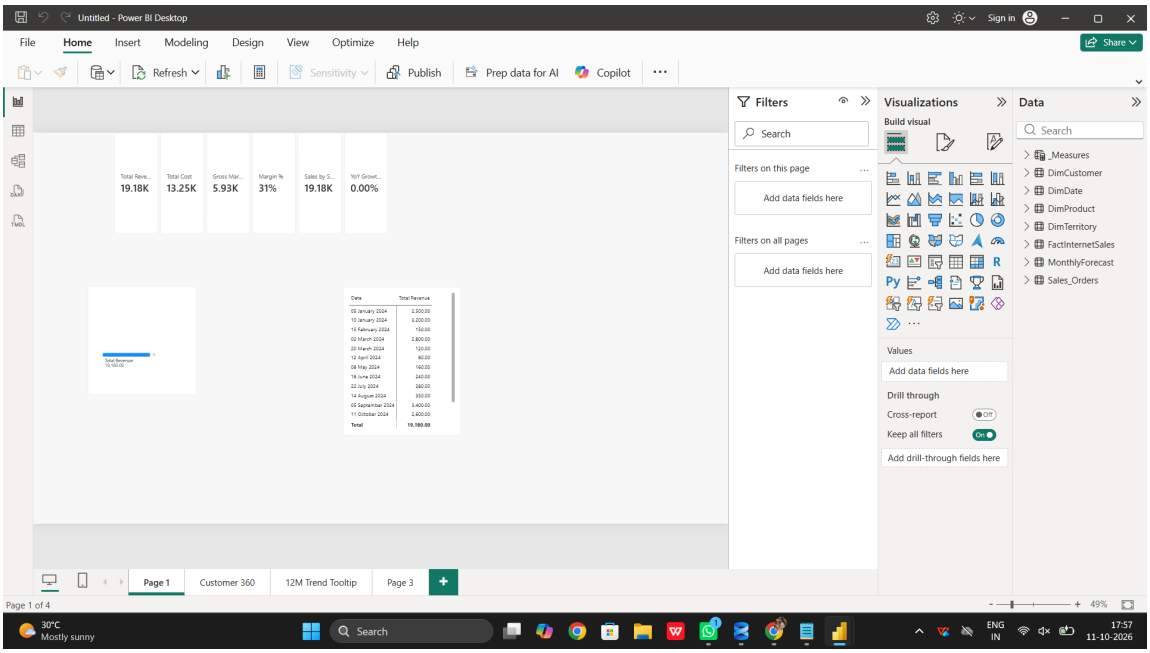


Figure 7.23: Visualizing Revenue, Cost, and Profitability Metrics

This image shows the sales performance dashboard developed in Power BI. It contains KPI cards for Total Revenue, Total Cost, Gross Margin, Margin Percentage, Sales by Ship Date, and Year-over-Year Growth. A revenue summary table and additional cards provide supporting details. These visuals consolidate important business metrics into a single report page, making it easier to review overall sales performance and profitability.

7.6.14 Exercise 4: Customer Churn Data Transformation

Caption: Assigning Appropriate Data Types to Churn Fields

This image displays the CustomerChurnPredictions table in Power Query Editor. The dataset contains CustomerKey, ChurnProbability, and SpendingCluster columns. Appropriate data types are assigned to the customer identifier, numerical churn probability, and categorical spending cluster. The column profiling indicators show that the displayed columns contain valid values without detected errors or empty entries. This preparation step ensures that the data is suitable for analysis and visualization.

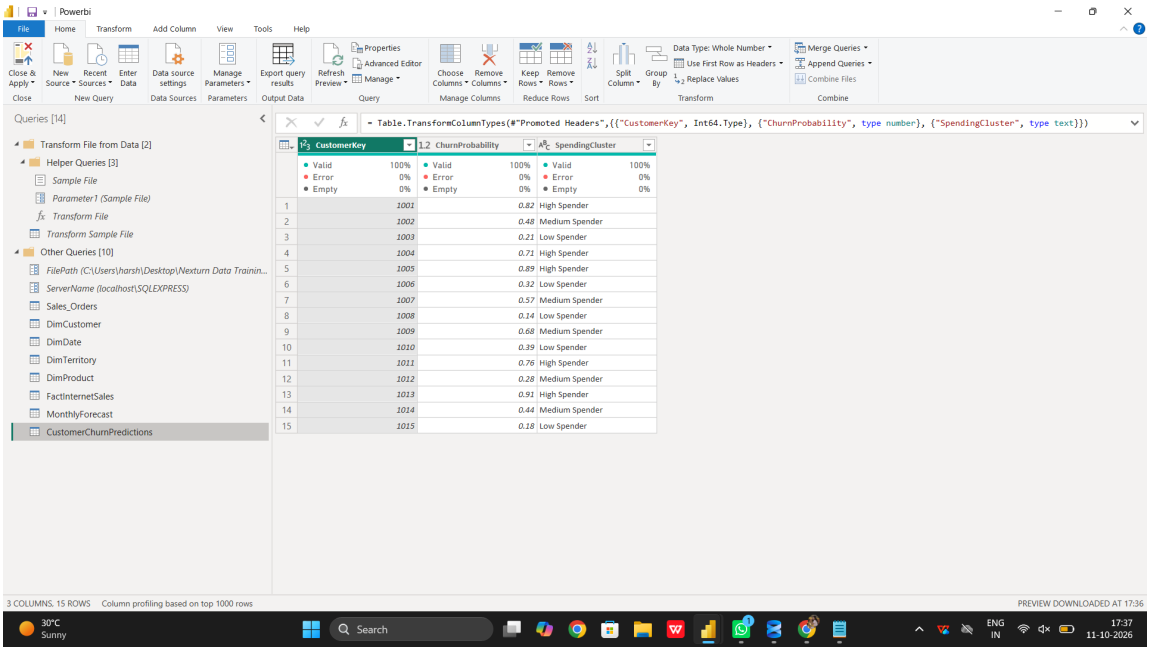


Figure 7.24: Preparing Customer Churn Data in Power Query

7.6.15 Exercise 5: Dashboard Design and Churn Risk Analysis

Caption: Customer Churn and Sales Analytics Dashboard

This image presents the combined Customer Churn and Sales Analytics Dashboard with a dark blue background and a prominent report title. It contains charts, tables, a donut chart, a churn risk tier slicer, and KPI cards displaying customer counts and average churn probabilities. The visuals compare spending clusters and churn risk categories, while the table provides selected sales transaction details. This integrated dashboard helps users explore customer segments and review churn-related patterns alongside sales information.

7.6.16 Exercise 6: SQL Database Inspection

Caption: Retrieving Database Table Names Using SQL Server

This image shows a SQL query executed in SQL Server Management Studio to retrieve the base tables available in the PowerBI_Engineering database. The query uses the INFORMATION_SCHEMA.TABLES view and filters the results to include only base tables, ordering them alphabetically by table name. The displayed results include DimCustomer, DimDate, DimProduct, DimTerritory, and FactInternetSales. This step verifies the database structure and identifies the tables available for the Power BI reporting workflow.

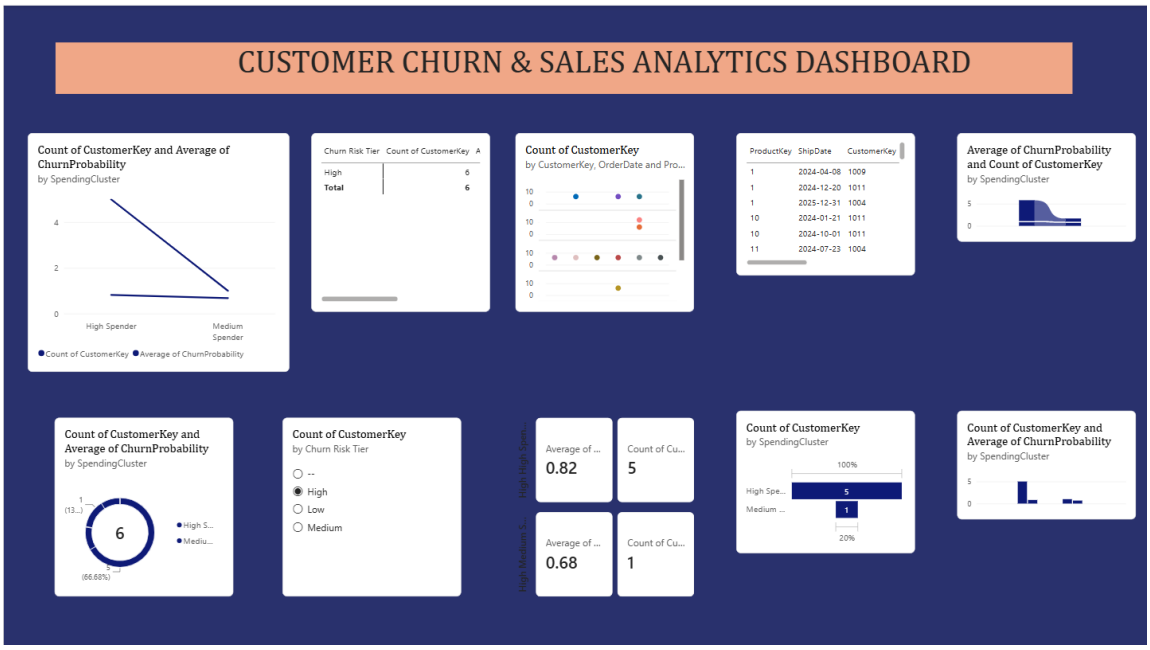


Figure 7.25: Combining Customer Churn and Sales Analytics

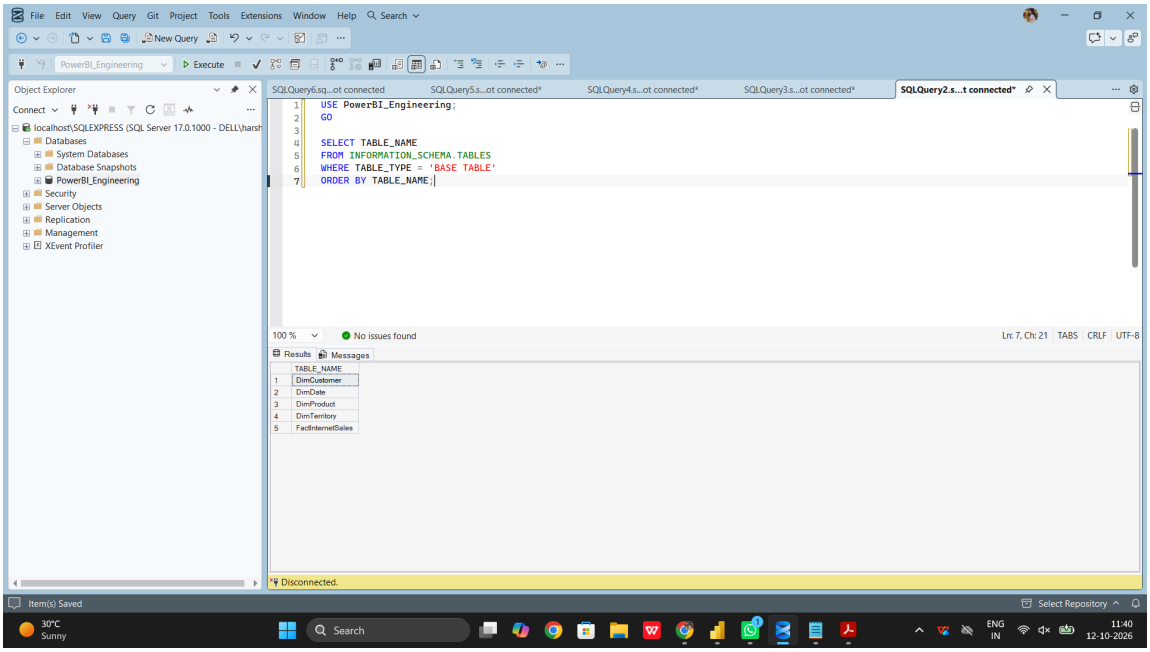


Figure 7.26: Listing Base Tables in the PowerBI_Engineering Database

7.6.17 Exercise 7: SQL Data Manipulation

Caption: Updating Customer Identifiers in the Sales Fact Table

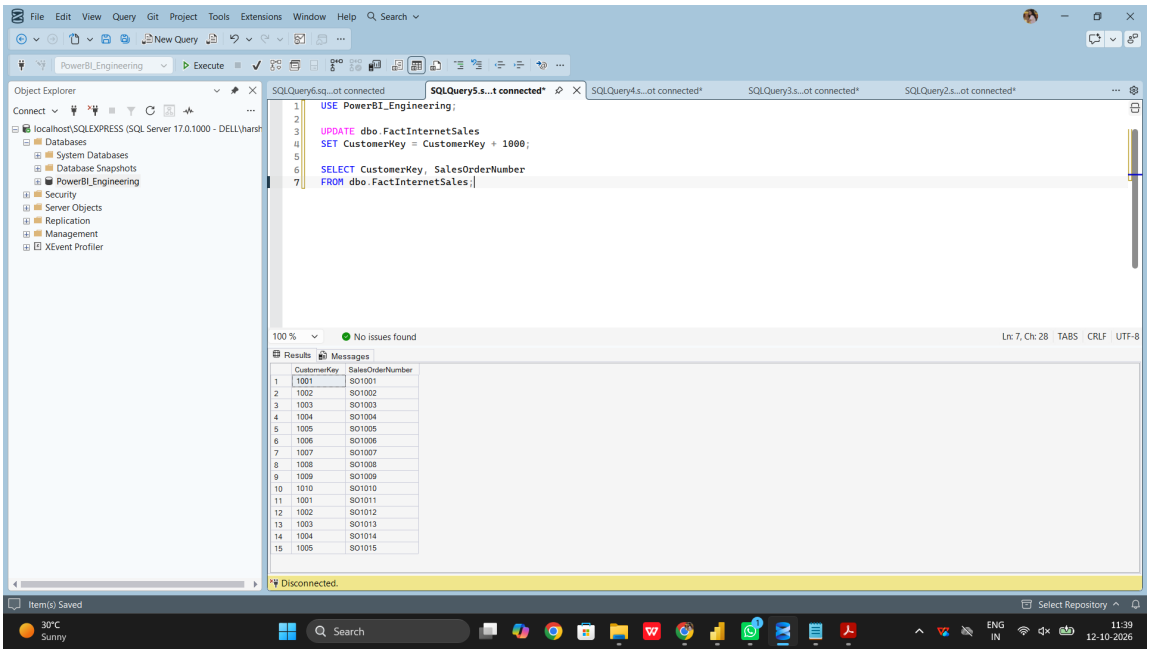


Figure 7.27: Executing an UPDATE Statement in SQL Server

This image demonstrates the use of the SQL UPDATE statement on the FactInternetSales table. The query modifies CustomerKey values by adding 1000 and then retrieves CustomerKey and SalesOrderNumber to display the results. It illustrates how SQL can be used to modify existing records and validate the resulting data. Such updates should be performed carefully because customer identifiers are used to establish relationships between fact and dimension tables.

7.6.18 Exercise 8: Date Dimension Generation

Caption: Populating the Date Dimension with Calendar Attributes

This image shows a SQL script that clears and repopulates the DimDate table with calendar information. The INSERT statement generates a date key using the YYYYMMDD format and derives the date, fiscal year, quarter, and month attributes. The results show 1,461 rows covering the period from January 1, 2023, through December 31, 2026. The date dimension supports consistent date-based filtering and time-series analysis in Power BI reports.

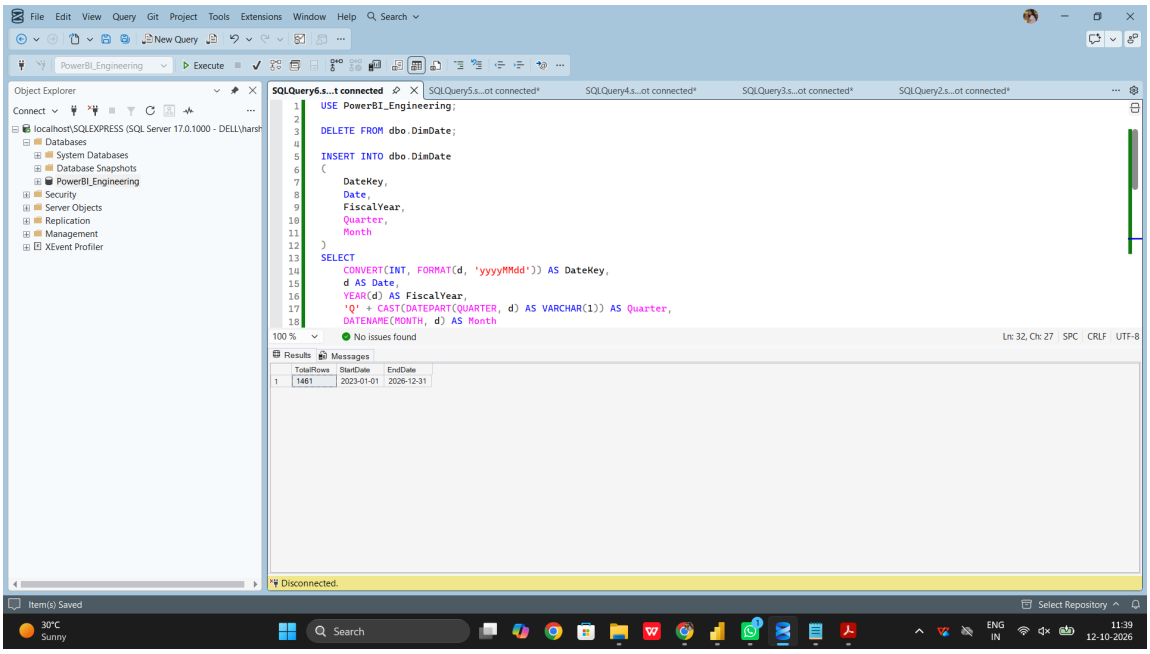


Figure 7.28: Generating Date Keys and Calendar Attributes

7.6.19 Exercise 9: DAX Calculated Column for Churn Classification

Caption: Creating Churn Risk Tiers Using DAX

This image demonstrates the creation of a calculated column named Churn Risk Tier using the DAX SWITCH and TRUE functions. Customers with a churn probability greater than 0.65 are classified as High risk, those with a probability greater than or equal to 0.35 are classified as Medium risk, and the remaining customers are classified as Low risk. The resulting column makes it easier to group customers into risk categories and build targeted visualizations for churn analysis.

7.6.20 Exercise 10: Final Customer Churn and Sales Reporting

Caption: Developing Interactive Customer Churn Visualizations

This image shows the customer churn analysis report with multiple visualizations arranged on a single page. It includes customer counts, average churn probabilities, spending cluster comparisons, a churn risk tier slicer, and a table of sales details. The interactive report layout allows users to explore customer categories and examine how churn risk is distributed across spending groups. It demonstrates the integration of SQL-based data preparation, Power Query transformations, DAX classification, and Power BI visualization in a unified analytics solution.

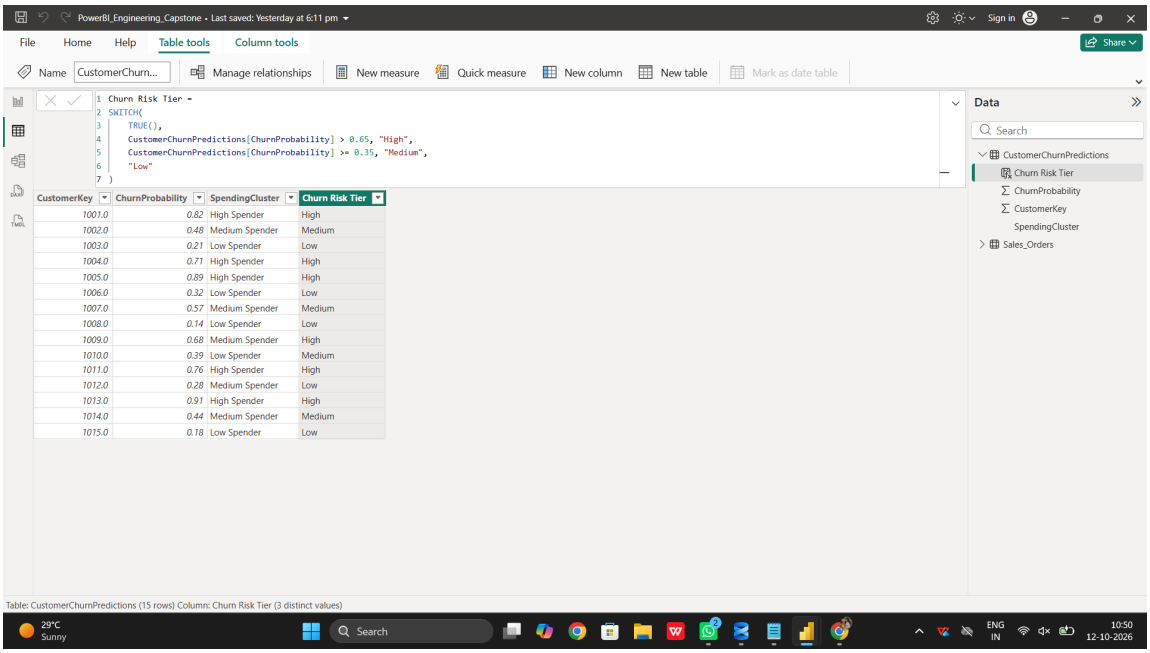


Figure 7.29: Classifying Customers into High, Medium, and Low Churn Risk Tiers

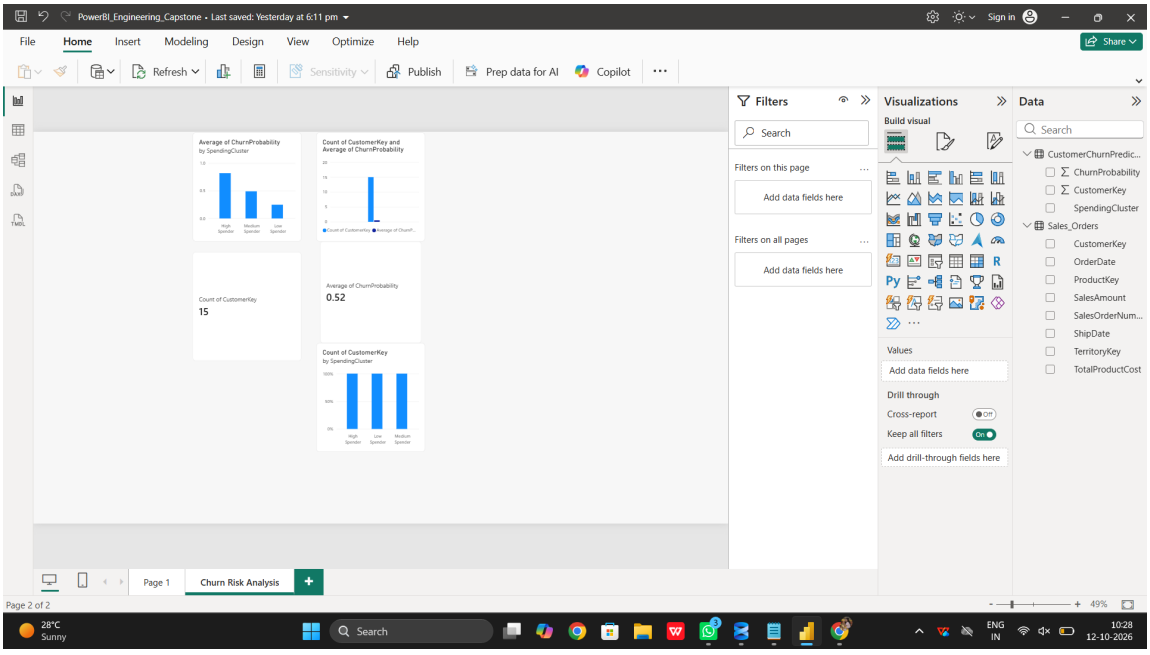


Figure 7.30: Exploring Customer Risk, Spending Behavior, and Sales Details

7.7 Conclusion

The Power BI Engineering project was completed by implementing all five exercises, covering the complete workflow from data ingestion and transformation to dimensional modelling, DAX calculations, interactive reporting, and deployment-related concepts. This project provided practical experience in working with Power BI Desktop, Power Query, SQL Server Management Studio, SQL Server, and DAX to prepare data and develop business intelligence solutions.

The first exercise focused on data ingestion and ETL operations, including connecting to different data sources, inspecting datasets, applying transformations, managing data types, and preparing data for analysis. The second exercise involved dimensional modelling, organizing fact and dimension tables, establishing relationships, and preparing the date dimension for time-based analysis. The third exercise focused on creating reusable DAX measures for revenue, cost, gross margin, margin percentage, year-to-date revenue, prior-year sales, year-over-year growth, rolling twelve-month sales, sales by ship date, and customer ranking.

The fourth exercise involved developing interactive Power BI reports and analytical visuals to present business information clearly. The fifth exercise covered deployment and operational-readiness concepts, including report publishing, refresh configuration, gateway requirements, and row-level security. Together, these exercises provided exposure to the major stages involved in developing a Power BI analytics solution.

7.7.1 Customer Churn and Sales Analytics Dashboard

As part of the reporting work, a Customer Churn and Sales Analytics Dashboard was developed to visualize customer behaviour and churn-related indicators. The dashboard includes visualizations for churn probability by spending cluster, customer counts by churn risk tier, customer-level details, and comparisons between spending clusters. It also includes KPI cards displaying average churn probability and customer counts, a donut chart, a bar chart, and a churn risk tier slicer for interactive filtering.

The dashboard brings multiple analytical views together on a single report page, making it easier to examine customer segments, compare high and medium spending clusters, and identify patterns in the available churn-related data. The combination of summary cards, charts, tables, and slicers supports a more interactive approach to exploring customer behaviour. The dashboard represents the practical application of Power BI visualization and reporting techniques to a customer analytics use case.

Overall, the completion of these five exercises strengthened my practical understanding of data preparation, SQL-based data handling, dimensional modelling, DAX, and inter-

active dashboard development. As a Data Engineering Trainee at Nexturn, this project helped me connect data engineering concepts with business intelligence and analytics requirements. It also provided valuable experience in organizing data, developing analytical calculations, validating outputs, and presenting information through a structured reporting solution.